



In vivo knee rotational stability 2 years after the ACL reconstruction using a quadriceps tendon graft with bone block and bone–patellar tendon–bone graft

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Abstract

Introduction The aim of this prospective randomised study was to evaluate clinical results and rotational stability at least 2 years after single-bundle anatomic anterior cruciate ligament reconstruction using a quadriceps tendon graft with bone block (BT) and bone–patellar tendon–bone graft (BTB).

Materials and methods In both groups (BT and BTB), 40 patients selected prospectively at random were evaluated. The mean follow-up after the surgery was 28 months (range 24–33 months). A navigation system was used to measure rotational stability of the knee joint. Cincinnati, Lysholm, and IKDC scores and visual analog score (VAS) were used to evaluate clinical results and the non-parametric Wilcoxon test was used for the statistical analysis.

Results After the BT reconstruction, the mean internal rotation of the tibia (IR) was 9.5°. In the contralateral healthy knee joint, IR was 8.6° at average. After the BTB reconstruction, the mean IR was 9.9°. In the contralateral healthy knee joint, IR was 8.7° at average. We did not find any statistically significant difference in IR stability between BT and BTB reconstruction. In terms of clinical results, regarding the VAS, patients perceive significantly more pain after the BTB reconstruction ($p < 0.05$). Kneeling was reported more difficult and painful after BTB reconstruction.

Conclusions The BT reconstruction of the ACL provides similar clinical results, less pain, better flexion and the same rotational stability of the knee in comparison with the BTB reconstruction.

Keywords ACL · Rotational stability · Quadriceps tendon graft

Introduction

The main purpose of ACL reconstruction is to restore the anterior–posterior and rotational stability of the tibia in its relation to the femur. The position of the femoral bone socket is crucial for knee kinematics and stability [1, 2]. The femoral foot-print position is well known [3–5]. The biomechanic studies describe the kinematics of the knee joint after different kinds of the ACL reconstruction [6–8]. Some researches deem the double-bundle (DB) reconstruction of the ACL to restore the rotational stability to a greater extent in comparison to the single-bundle (SB) technique [9, 10]. However, the DB technique has disadvantages such as more complications during the surgery and higher cost [11]. In these studies, hamstrings tendons or bone–patellar tendon–bone (BTB) graft were used. There is an absence of studies discussing the stability of the knee joint after the reconstruction of the ACL using the quadriceps bone-tendon graft (BT graft).

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The objective of this prospective randomised study was to evaluate knee rotational stability at least 2 years after the anatomic single-bundle (SB) ACL reconstruction with the BTB and BT grafts in comparison to the contralateral healthy knee joint. The anterior knee pain and the possibility of the kneeling were analysed, as well.

Two hypotheses were postulated: (1) The BT reconstruction restores the knee stability in internal rotation (IR) without any significant difference in comparison to the BTB reconstruction. (2) The knee after the BT reconstruction is less painful than after the BTB reconstruction.

Materials and methods

80 patients with an average age of 30.4 years (16–52 years; 44 men/36 women; 45 right/35 left knees) underwent ACL reconstruction because of isolated ACL tear between 2015 and 2017 (Table 1). The ACL injury was diagnosed using magnetic resonance imaging (MRI). After prior careful instruction and signing written informed consent patients were enrolled in the study. The study was approved by the Ethics Committee and by the review board. The inclusion criteria were strict: (a) isolated ACL lesion; (b) no injury or surgery done on knee joints before; (c) age of 15–55 years; (d) body mass index (BMI) below 30. All other patients with associated injuries (meniscal lesions, fractures, collateral ligament ruptures) or with systemic diseases were not included. The cases with partially torn ACL confirmed by arthroscopy were not included either.

In BT group, the anatomic SB reconstruction of the ACL with the BT graft from the quadriceps muscle was done (40 patients). In the BTB group, 40 patients received the anatomic SB reconstruction of the ACL with BTB graft from the ligamentum patellae. The choice of patients for one of the two above-mentioned techniques of the ACL reconstruction was done randomly using the permuted block randomization by Random Number Generator Software 7.0. To detect a difference of 1° in absolute values during the movement of the tibia against the femur, 80 subjects were necessary to analysed.

Patients completed the Cincinnati, Lysholm and IKDC scores at least 24 months after the surgery. They completed

the Visual Analogue Score (VAS) for the analysis of the anterior knee pain and the possibility for kneeling, too.

Surgical technique

After the patient had been introduced into anaesthesia, a diagnostic arthroscopy confirmed the complete rupture of the ACL and excluded injuries of other intra-articular structures.

To perform the anatomic SB reconstruction, the BT graft was harvested from the quadriceps tendon and the upper part of the patella through the transverse skin incision of 4 cm in length. The specialized instrumentation was used to harvest it (Figs. 1, 2).

The BT graft was prepared to the length of 9 cm and width of 9 mm. For the BTB reconstruction, the ligamentum patellae BTB graft was harvested from the longitudinal skin incision. The bone from blocks from the patella and from the tibia measured 25 × 10 mm. The tendon had 9 mm in width. In both groups, the femoral tunnel was drilled in the anatomic footprint of the ACL through the anteromedial portal. The tibial tunnel was drilled into the centre of the ACL original footprint. Grafts were fixed with the biocomposite interference screws.

Experimental system

The optical computer navigation system OrthoPilot® (Aesculap, Tuttlingen, Germany) was used to evaluate stability at the follow-up control. All measurements were done on both the reconstructed and healthy knee. Tripods composed of 3 passives markers were fixed 120 mm above and under the centre of the patella with firm elastic stripes (Fig. 3).



Fig. 1 Skin incision for the harvest of the BT graft

Table 1 Patient demographics data

	BT	BTB
Women	19	17
Men	21	23
Right knee	22	23
Left knee	18	17
Average age (years)	29.8	31.0



Fig. 2 Double knife for harvest of the BT graft



Fig. 3 The position of the passive markers fixed to the patient's leg

Measurement started in standing position with both feet separated by 30 cm in 30° weight-bearing flexion. The patients were asked to perform the maximal external trunk rotation to develop the internal rotation of the tibia against the femur. All measurements were repeated three times for each knee joint. The degree of flexion and rotation was followed on the screen of the navigation system.

Statistical analysis

All data were statistically treated by STATISTICA 9.0 software. The description of the deviation of movement in degrees included mean, standard deviation and range for continuous variables. For determination of the statistically significant differences between the mean values, the non-parametric Wilcoxon test was used. The p value < 0.05 was considered statistically significant.

Table 2 The average values of Cincinnati, Lysholm and IKDC scores at least 24 months after the BT and BTB anatomic SB reconstruction of the ACL

	BT	BTB	p value
Cincinnati	89.6 (83–100; SO 6.4)	95.1 (83–100; SO 7.2)	0.23
Lysholm	90.1 (76–100; SO 6.7)	93.4 (64–100; SO 9.6)	0.47
IKDC	78.3 (65.5–93.1; SO 10.5)	74.2 (55–80; SO 10.2)	0.13

P value shows no statistical difference between both techniques at least 24 months after the operation

Table 3 The average values of VAS score (focused on anterior knee pain) at least 24 months after BT and BTB reconstruction of the ACL

	BT	BTB	p value
VAS – anterior knee pain	1.4 (0–3; SO 1.7)	2.8 (1–5; SO 1.2)	0.03

p value shows the statistical difference between BT and BTB techniques at least 24 months after the surgery

Source of funding

There were no financial resources used for this research.

Results

The mean follow-up was 28 months (range 24–33 months). All patients finished the rehabilitation program 6 months after the reconstruction of the ACL.

After the BT reconstruction, the mean internal rotation of the tibia (IR) was 9.5° (range 5°–14°, SD = 3.5°). In the contralateral healthy knee joint, IR was 8.6° at average (range 5°–15°, SD = 4.3°). We found statistically significant difference in IR stability between reconstructed and healthy knees ($p = 0.036$). After the BTB reconstruction, the mean IR was 9.9° (range 4°–17°, SD = 4.7°). In the contralateral healthy knee joint, IR was 8.7° at average (range 5°–15°, SD = 3.9°). We found statistically significant difference in IR stability between reconstructed and healthy knees ($p < 0.027$). We did not find any statistically significant difference in IR stability between BT and BTB reconstruction ($p < 0.134$).

In terms of clinical results (Cincinnati, Lysholm, IKDC), we did not find any statistically significant difference between BT and BTB group (Table 2).

Regarding the VAS focused on anterior knee pain, patients perceive significantly more pain after the BTB reconstruction ($p = 0.03$) (Table 3).

Kneeling was reported more difficult after BTB reconstruction (Table 4).

Table 4 The average values of VAS score (focused on kneeling) at least 24 months after BT and BTB reconstruction of the ACL

	BT	BTB	<i>p</i> value
VAS—kneeling	2.5 (1–4; SO 1.2)	6.1 (2–10; SO 2.5)	0.01

p value shows the statistical difference between BT and BTB techniques at least 24 months after the surgery

Discussion

The key finding of this study is that the BT reconstruction of the ACL provides the same rotational stability improvement of the knee in comparison with the BTB reconstruction. But the rotational stability after respective procedures does not improve to the point of a healthy unoperated knee.

There are very few studies dealing with the objective measurement of rotational stability of the knee in vivo two or more years after the ACL reconstruction even though the remaining rotational laxity could be the main source of pain. Several studies demonstrate higher efficacy of the DB reconstruction of the ACL on the restoration of the rotational stability in comparison to the standard SB technique, in an experimental setting and in vivo, as well [10, 11]. In majority of these studies, hamstrings tendons have been used. In our study, the rotational stability of the knee joint was analysed to compare the quadriceps and the ligamentum patellae grafts for the purpose of ACL reconstruction.

One of the main issues is how to objectively measure rotational stability of the knee joint. Ho et al. have presented the study where they describe a navigation system used to measure the rotational stability of the knee joint in fresh frozen cadavers [7]. Hoshino et al. used dynamic stereo X-rays to make radiographic images during the downhill treadmill running in comparison with the contralateral knee joint [12]. The usage of the navigation system to evaluate the laxity of the knee joint was described by Chouteau et al. [13]. They used the navigation to control the laxity of the knee joint after the partial reconstruction of the ACL in comparison to the contralaterally healthy knee joint. Gobbi et al. [14] demonstrate no difference in the rotational laxity between the SB and DB reconstruction in their study 3 years after the operation. Similar results were published by Suomalainen et al. 5 years postoperatively [15].

The BTB graft is considered to be firmer than the hamstring graft, with a lower percentage of failure [16]. There are several analyses which describe the ACL reconstruction with BTB autografts to be superior in restoring the rotation stability of the knee joint and that patients return to higher levels of activity in comparison with patients undergoing operation using the hamstring autografts [17]. To the best of our knowledge, there is no study comparing BTB and BT grafts in terms of the rotational stability of the knee joint.

BT graft is still used especially for the revision surgery, but it could be successfully used for the primary ACL reconstruction, as well [18]. Barié et al. describe in their long-term follow-up study good results after primary ACL reconstructions using different autografts 7.5 years after the surgery [19]. This study compares rotational stability after the BTB reconstruction with the quadriceps BT one. In 2016, Lee et al. describe in their study similar knee stability and functional outcome scores after using hamstrings autografts [20]. In the meta-analysis of Slone et al. from 2015 [21], 14 studies were included, including 1 154 ACL reconstructions with quadriceps BT autograft. Six studies directly compared quadriceps BT autografts with BTB autografts. Stability outcomes (Lachman, pivot-shift, and instrumented anterior laxity testing), functional outcomes (IKDC, Lysholm), overall patient satisfaction, range of motion, and complications were similar.

In a study done by Hunnicutt et al., the function of the quadriceps muscle after the BT and BTB ACL reconstruction was evaluated [22]. The function of the knee joint was analysed by the IKDC questionnaire. Authors concluded that objective function and subjective results were without any statistical difference. Patients with quadriceps BT autografts demonstrated similar short-term quadriceps recovery and postsurgical outcomes just as patients with BTB autografts did. Although the measurements were done only 8 months after the surgery (on average).

The function of the knee joint is a very wide scientific topic. There are several studies that analysed function of the knee joint after various types of the ACL reconstruction [23]. Surrounding muscles have quite considerable impact on rotational stability of the knee joint. Both the reconstruction of the ACL and good function of surrounding muscles improve the function of the knee joint during weight bearing and sport activities [24].

In the study of Hunnicutt, less donor-site morbidity was seen in patients who underwent quadriceps BT ACL reconstructions [22]. The possibility of better kneeling down is another benefit of the BT graft.

The strength of our study is the usage of a special technical device to objectively evaluate difference of rotational stability in weight-bearing condition of contralateral healthy knee joints and knees with ACL reconstruction. The 2-year follow-up allows us to evaluate the rotational kinematics of the knee joint during the normal weight bearing.

Our study certainly had its limitations. Soft tissue with attached markers could move against the bones (although the same amount of movement would happen in healthy knees) and the reliability and repeatability our kinematic evaluations is to be considered.

In conclusion, our newfound data confirmed both of our hypotheses. The BT reconstruction of the ACL provides similar clinical results, less pain, easier kneeling and the

same rotational stability of the knee in comparison with the BTB reconstruction. Rotational stability after both procedures is below that of a healthy knee.

Declarations

Conflict of interest There is no conflict of interest.

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